# Thesis checklist

Open issues only, reassessed against the current source. Line numbers may shift.

## 3.5.2 Lemma 7

- □ **Complete the Gaussian-kernel sentence.** The sentence after the lemma ends with “because ...”. Show directly that L(x) = (2π)^(-1/2)(1-x²)exp(-x²/2) satisfies the stated decay bound. The revised comparison with square integrability does not claim the converse. [chapter 567, 742]

## 3.2.2 Bregman and beta-divergences

- □ **Repair the beta-divergence domain.** The real power t^(1+beta) is written for all real t and arbitrary real beta > 0; reconcile the domain and regularity assumptions, including the separate quadratic case. [chapter 175-183]
- □ **Match the cited symmetry theorem.** The stated application cites Theorem 12 but does not state its Legendre-type hypothesis. Verify the assumptions for the functional or supply a separate proof. [chapter 159-163; arXiv:2604.24047, Theorem 12]
- □ **Correct the scalar test functions.** The proof of affine invariance uses phi(x) after defining p_t, while the symmetry proof uses p(x)=x on [0,1] to conclude a formula for every scalar. Correct the variable and justify the scalar conclusion. [chapter 169-172]

## 3.5.2 Proposition 8

- □ **Handle r = 0 in displayed formulas.** The statement includes r = 0, but the transformed-distance identity and endpoint expression contain 1/r. Give their logarithmic form or explicitly take the limit. [chapter 571-582, 675]
- □ **Establish positivity before evaluating the integral.** Q-hat_r(h_0) is proved positive after the Gaussian integral is evaluated. Move the existing positivity argument before the square-root formula. [chapter 699-708]
- □ **Correct the Laplace-method remark.** The cited source is blank and the displayed integral and square-root formula need correction, including the sign of the second derivative. State suitable hypotheses for a nondegenerate interior maximum. [chapter 712-717]
- □ **Correct the energy interpretation.** The text identifies Q-hat_r with the squared derivative norm but its definition contains the factor (1/2)h^[2(1-r)]. Qualify the finite-N row-sum interpretation using the sequence-level result. [chapter 576, 726]
- □ **Resolve the empty appendix heading.** The proof of Proposition 8 is in the main chapter, while the corresponding appendix subsection has no proof. Resolve the heading and any cross-reference. [chapter 579-710; appendix 371]

### 3.5.2 Proposition 9: appendix proof

- □ **Introduce notation before it is used.** Define the transformed coordinates t_i and endpoints, m(t), q_0, sigma_N, and the relevant maximizing-index sequence. Keep N and subsequence indices consistent. [appendix 373-404]
- □ **Justify the boundary row-sum bound.** The uniform upper-bound argument allows its limiting point to be an endpoint but refers to Proposition 8, whose statement assumes an interior limit. Explain the boundary case separately. [appendix 382-401; chapter 571-578]
- □ **Correct the Rayleigh-bound notation.** Use b_N consistently in the row-sum minimum and identify the correct index set for the minimum. In the pair-count explanation, j is outside I_N; up to k indices can occur on each side. [appendix 421-430]
- □ **Correct the sign in the E_N bound.** The displayed lower bound for sigma_N #I_N contains u-v, although u<v. Use the interval length v-u and check the resulting limit. [appendix 432]
- □ **Connect the final index to the stated maximizer.** The last step switches to i_N without identifying it with the maximizing index used at the start and with k_N in the proposition. Make that identification and limit argument explicit. [appendix 375-380, 435-437; chapter 732-734]

## Final check

- □ **Compile after revisions.** Check figures, captions, labels, bibliography, cross-references, and remaining marked TODOs in the compiled thesis. [thesis source and compiled PDF]

**Reassessment notes.** The local remainder now uses a bound independent of j; the duplicate local expansion and notation errors previously listed have been fixed. The rescaling and tail calculation is already shown. A prior checklist incorrectly characterized the revised kernel-decay sentence; the outstanding item is its unfinished Gaussian explanation.